Artifact-Aware Explainable Deepfake Detection via Fake-Only High-Frequency Alignment Supervision

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***Abstract*—In recent years, deepfake generation have advanced significantly that has increased the fidelity and accessibility of manipulated AI made content. Although current deepfake detection models achieve near perfect accuracy, but that does not entirely indicate that it attends high-frequency forensic artifacts. And this raise concerns about model's reliability and validity, which limits clarity trustworthiness in real-world deployment.**

This paper investigates the forensic consciousness of convolution neural networks (CNNs) in deepfake detection. A baseline EfficientNet-B0 model is trained, and Grad-CAM is employed to examine attention visualization. It detects the spatial attention patterns. FFT (Fast Fourier Transform) is used to extract high-frequency artifact maps, and cosine similarity is computed to measure alignment between model attention and artifact regions.

The baseline model reveals minimal alignment, with a gap of 0.013 between real and fake samples. To enhance artifact sensitivity, we propose a fake-only high-frequency alignment loss that encourages attention towards forensic artifacts aware learning. Experimental outcomes states that the proposed supervision significantly increases the alignment gap to 0.262 while preserving 99.96% validation accuracy. These findings highlight the need to incorporate artifact aware supervision to improve interpretability without degrading model performance.

The extension of the present study will be applying the artifact aware supervision framework to diverse cross-domain datasets and to the detection of AI-generated content in videos.

I. Introduction

The emerging manipulated content generation models such as Generative Adversarial Networks (GANs) [1], [2] and diffusion architectures, have enabled the growth of production in synthetic media. Among these, Deepfakes,in particular, creates a significant concern in society by enabling very realistic facial manipulation in images and videos. Although these technologies have creative and entertainment applications, they also possess serious societal risks such as misinformation, identical issues, digital fraud, etc.

To overcome these threats, deep learning-based detection methods have been vividly studied. For example, Convolution Neural Networks, specially pretrained models such as EfficientNet [3], demonstrate impressive accuracy through transfer learning. However, most of the methods that exist often comes with a critical question: are these models particularly learning from relevant forensic artifacts ? High performance does not always promise frequency oriented artifact-aware learning, as it may instead utilize specific biases and not the genuine traces.

Prior findings suggest that generative models introduce fine frequency-domain irregularities like high frequency artifacts, abnormal texture patterns and unusual noise distributions. These inconsistencies are often taken no notice of in the spatial domain but can be disclosed ...through frequency analysis using FFT [7], [8]. By converting the images into frequency signals, it becomes possible to track components those serve as good artifact indicators.

Despite this understanding, limited research is done on the fact that whether CNN-based models attend to such artifacts and the concern remains underexplored. Moreover, explainability is a major challenge. Although visualization models like Grad-CAM [4] shows attention regions, it does not necessarily mean that the highlighted area correspond to meaningful artifact-aware evidence.

In this work, to address this gap, we propose an explainable deepfake detection framework that is also artifact aware, as illustrated in Fig. 1. It calculates the relationship between frequency-domain and attention maps. FFT extract high-frequency maps. The alignment gap between attention maps and high frequency regions are quantitatively measured using cosine similarity.

Our experimental results reveal that while baseline model achieve near-perfect classification accuracy, it exhibits limited attention to forensic artifact regions. To improve this, we introduce a fake-only high-frequency aligned framework that guides attention towards artifact regions in samples. This approach enhance clarity and understanding while preserving strong performance to promote reliable and explainable deepfake detection technique.

II. Related Work

Deepfake detection has been extensively explored through Convolution neural networks [6], [11], vision transforms and other hybrid frameworks. Primitive approaches were more focused on spatial feature inconsistencies such as color inconsistencies, blending artifacts and uneven texture distributions in face manipulation detection methods [5], [6]. Pretrained architecture models such as EfficientNet have shown strong performance through transfer learning.

More recently, studies identified irregularities in the frequency domain [7], [8], particularly texture and noise patterns. Fourier-based analysis are implemented to exploit these signals and promote frequency-aware detection methods. Despite these facts, existing studies primarily focus on maximizing classification accuracy, often overlooking whether the model corresponds to meaningful artifacts.

To improve clarity and trust, explainable AI methods such as Grad-CAM [4] have been applied to investigate model attention patterns. These techniques provide qualitative insights and quantitative evaluation of attention alignment with forensic artifact remains underexplored.

To address this limitation, the approach in this paper measures and strengthen the alignment between attention maps and frequency artifact regions using a fake-only alignment supervision, hence promoting artifact aware deepfake detection technique.

The authors in [14] proposed an ensemble-based deep learning architecture for the detection of deepfake and also highlighted key research challenges.

Other approaches include residual learning [9], multi-task learning frameworks [12], and generative model fingerprinting techniques [13].

III. Proposed Methodology

The overall pipeline of the proposed artifact-aware explainable deepfake detection framework is depicted in Fig. 1.

A. Baseline Deepfake Detection Model

Here, EfficientNet-B0 [3] is employed as the backbone architecture for its strong performance. The model is fine-tuned for binary classification (Real vs Fake) and is initialized with ImageNet pretrained weights.

Input images are preprocessed where they are resized to 224×224 pixel, followed by normalization before being passed to the convolution network. The final layer is modified to two output classes. For classification upgradation, cross entropy loss is used.

B. Explainability using Grad-CAM

To interpret the internal reasoning of the model, Grad-CAM is used in the final fully connected layer of the network. This generates attention maps that help in identifying spatial regions contributing most significantly to model's performance.

Such visualization promote qualitative insights and examine whether the framework attends significantly to meaningful features.

C. Frequency-Domain Artifact Extraction

The input image is transformed into frequency domain with Fast Fourier Transform (FFT) [7], to analyse spectral components, that may reveal the inconsistencies that the synthetic content possess.

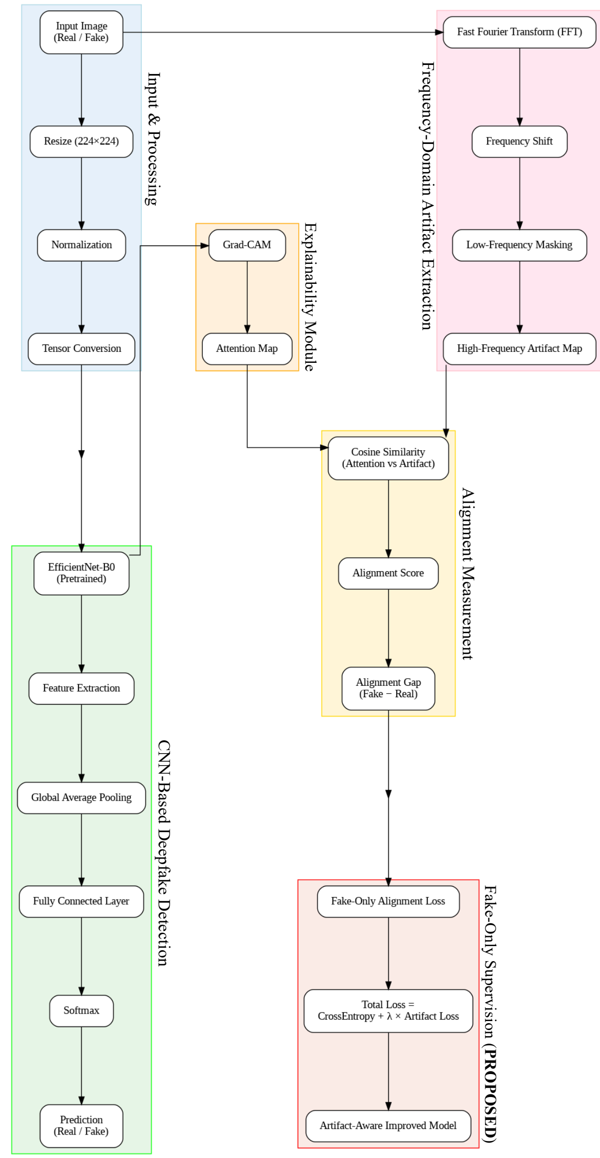

Fig. 1. Proposed Artifact-Aware Explainable Deepfake Detection Framework.

The extraction method involves:

- Reducing channel redundancy by converting the input image into greyscale.
- Application of a 2D FFT.
- Frequency shifting, to center low-frequency components.
- Using central masking to suppress low-frequency (smooth) information.
- Extraction of high-frequency maps.

D. Attention-Artifact Alignment Measurement

To evaluate the correspondence between attention and artifact regions, cosine similarity is computed between the high-frequency artifact maps and the corresponding Grad-CAM attention maps.

The metric is computed as:

$$\text{Cosine Similarity} = \frac{A \cdot F}{\|A\|\|F\|} \tag{1}$$

where A denotes model attention and F denotes artifact maps.

To evaluate biased behaviour, alignment gap is further computed, which is defined as:

$$\text{Alignment Gap} = \text{Fake Alignment} - \text{Real Alignment} \tag{2}$$

Stronger artifact-consistent learning is reflected through a larger alignment gap.

E. Fake-Only High-Frequency Alignment

An additional fake-only alignment loss is introduced to promote artifact focused learning. It is defined as:

- Fake samples: maximize the attention-artifact alignment
- Real samples: no alignment constraints

The optimization formula is defined as:

$$\mathcal{L}_{total} = \mathcal{L}_{CE} + \lambda \mathcal{L}_{align}^{fake} \tag{3}$$

where $\mathcal{L}_{CE}$ and $\mathcal{L}_{align}^{fake}$ denotes classification cross-entropy loss and cosine alignment loss, λ is the hyperparameter.

IV. Experimental Setup

A. Dataset

The experiments were conducted using the (RealvsFake 162k by Wish dataset), similar to large-scale deepfake datasets such as DFDC [10], that contains:

- 81,000 real images
- 80,972 fake images

But a balanced subset was created for more efficient training:

- 32,000 training samples
- 8,000 validation samples

Input images are resized to 224×224 resolution.

B. Implementation

- EfficientNet-B0 backbone architecture
- Adam optimizer
- Cross-Entropy loss function
- T4 GPU (Colab) as training device
- 5 training epochs, 32 batch size
- Grad-CAM in the final convolutional layer
- FFT based artifact extraction.

V. Results and Analysis

A. Classification Performance

TABLE I
Classification Performance

Model	Accuracy (%)	Precision	Recall	F1-Score
Baseline EfficientNet	99.91	1.00	1.00	1.00
Proposed Artifact-Aware	**99.96**	1.00	1.00	1.00

The classification performance comparison between the baseline and proposed models is presented in Table I. The result reveals that both baseline and proposed model achieve good classification accuracy, while the artifact-aware framework improves interpretability without compromising performance.

B. Attention–Artifact Alignment

TABLE II
Attention–Artifact Alignment

Model	Fake Alignment	Real Alignment	Alignment Gap
Baseline	0.805	0.792	0.013
Proposed (Fake-Only HF)	**0.960**	0.699	**0.262**

Alignment analysis in Table II states that EfficientNet shows limited attention towards high-frequency artifact regions. Contrastingly, the proposed methodology increases the alignment gap showing improved learning.

C. Qualitative Visualization

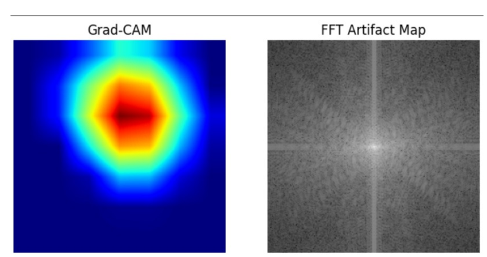

Fig. 2. Grad-CAM Attention and FFT Artifact Map Comparison

As shown in Fig. 2, the proposed model focuses more effectively on high-frequency artifact regions compared to the baseline. Grad-CAM visualization strengthen quantitative results.

D. Alignment Gap Improvement

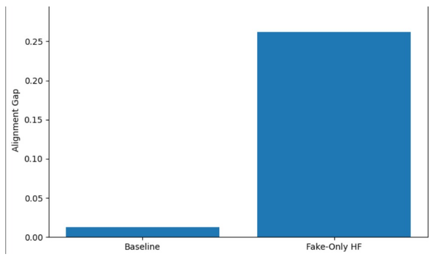

Fig. 3. Discriminative Alignment Gap Improvement

Fig. 3 illustrates the improvement in alignment gap achieved by the proposed method. The significant increase in alignment gap confirms that the proposed fake-only supervision model successfully enhances high-frequency artifact map learning without degrading overall model performance.

E. Ablation Study

TABLE III
ABLATION STUDY

Supervision Type	Accuracy (%)	Alignment Gap
No Supervision	99.91	0.013
High-Frequency (All Samples)	99.93	0.007
Fake-Only High-Frequency	**99.96**	**0.262**

The ablation study results shown in Table III indicate that applying supervision to just fake samples produce most discriminative improvement.

VI. CONCLUSION

This work investigates whether models that are high performing can achieve near-perfect accuracy while genuinely learning meaningful artifacts or just depends on low frequency smooth regions. The baseline model achieved 99.91% validation accuracy, but it has no or limited alignment between attention maps and artifact regions.

To overcome this gap, a fake-only alignment supervision framework is designed to encourage artifact-aware learning.

The research findings reveal improvement in alignment gap without sacrificing the model's classification performance.

Despite of the strong performance, this study has several limitations. It is evaluated on a limited dataset and primarily address image based inputs.

Further work will focus on integration of transformer-based detection methodology and extension to video-level analysis for more generalization to real-world scenarios.

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